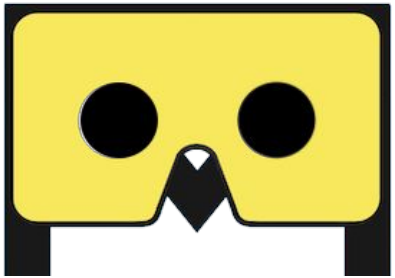
Web AR Workarounds

To make web based AR work on all Browsers with one build



8th Wall
owned by Niantic Spatial, Inc.

Maintained by:
Niantic Spatial, Inc.



Mind.AR
Open Source, MIT License

Maintained by:
A single individual developer, going by "hiukim" on GitHub.



AR.js
owned by Niantic Spatial, Inc.

Maintained by:
The AR.js org, A GitHub Organisation

World Tracking (SLAM)

SLAM = Simultaneous Localisation And Mapping

Possible with: 8th Wall

How?
Uses the camera feed (Tracking Feature Points) to create a point cloud

Uses the Gyroscope (Rotation) and Accelerometer (Gravity & acceleration direction) to measure users Movement in 6 DOF



Note: this example is App based but can be reproduced on web using 8th Wall. There is a web based version accessible for some furniture in the online Ikea store

▸ Video: https://www.youtube.com/watch?v=UudV1VdFtuQ

World Tracking (SLAM) - Additionals

Possible with: 8th Wall

- **Plane/Surface Detection**
  Recognize a ground plane in space
- **Hit Test**
  Tap on 2D screen, place in 3D space
- **Lighting Estimation**
  Estimate brightness and color of the ambient light
- **World Scale**
  Content At Real World Scale



Note: this example is App based but can be reproduced on web using 8th Wall. There is a web based version accessible for some furniture in the online Ikea store

▸ Video: https://www.youtube.com/watch?v=UudV1VdFtuQ

Sky Effects

Possible with: 8th Wall

How?

- Segmentation: A neural network classifies each camera pixel as "sky" or "not sky"
- Mask: A black and White Mask is created that defines where your Texture or objects are rendered



▸ Video: https://www.youtube.com/watch?v=i41_ZaUMLvc

Face Tracking

Possible with: 8th Wall, Mind.AR

How?
Using the camera feed, a machine-learning model finds a face in the and predicts a 3D mesh of landmarks on it. 3D content is attached to that mesh.



▸ Video: https://www.youtube.com/watch?v=xNG_2grmv-s

-> MindAR Live Demo

Location Based

Possible with: AR.js

How?

- To get user position:
  Uses GPS (longitude/ latitude)
- For Viewing direction:
  Compass and orientation sensors (gyroscope, accelerometer, magnetometer)
- Objects scaling: virtual objects can can be scaled by distance to user



Note: It is genuinely difficult to find good video examples of this tracking Option. So here is a location test tt50xi uploaded to YouTube.

▸ Video: https://www.youtube.com/watch?v=biCKHVefIe8

Location Based

ISSUES

- GPS is only 3-10m accurate
- compass becomes unreliable inside buildings or near metal
- No internal SLAM (but maybe it can be combined?)

-> Needs to be used on big open forgiving outdoor planes



Note: It is genuinely difficult to find good video examples of this tracking Option. So here is a location test tt50xi uploaded to YouTube.

▸ Video: https://www.youtube.com/watch?v=biCKHVefIe8

Fun Fact! *Pokémon Go worked with GPS and local SLAM. Its player base became the data source for Niantics visual map (VPS). According to Niantic, combining that VPS with SLAM can place digital content in the real world with centimeter accuracy.*

Marker Tracking

Possible with: AR.js

How?

- Markers are designed to be easy for computers
- Simple black and white squares, thick borders for recognition, image inside border to tell the markers apart
- 3D content is placed relative to the markers



▸ Video: https://www.youtube.com/watch?v=hdXY1tFQ2Hw

! Use markers when robustness, speed or many IDs matter more than looks

Image Tracking

Possible with: MindAR, AR.js, 8th Wall

How?

- recognizes a known 2D image in the camera feed and calculates its position and rotation relative to the camera
- 3D content is placed relative to the image



Note: This Video Shows specifically curved image markers, an 8th Wall only feature

▸ Video: https://www.youtube.com/watch?v=COOAcJJYnCQ

What Makes A Good Target Image?



8th Wall's good/bad image target example

- Varied and Abundant Detail
- High Image Contrast (think black & white)
- sharp corners & lines



- No repetitive Patterns
- No excessive Blank space
- Non-reflective Surfaces